

EC200U&EG91xU Series QuecOpen(SDK) Ping API Reference Manual

LTE Standard Module Series

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About the Document

Revision History

Version	Date	Author	Description
-	2021-11-05	Herry GENG	Creation of the document
1.0	2021-11-25	Herry GENG	First official release
1.1	2023-09-08	Lambert ZHAO	- Updated the document name based on the unified QuecOpen naming. - Added applicable modules EG912U-GL and EG915U series.

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1 Introduction

Quectel EC200U series and EG91xU family modules support QuecOpen® solution. QuecOpen® is an embedded development platform based on RTOS, which is intended to simplify the design and development of IoT applications. For more information on QuecOpen®, see **document [1]**.

This document is applicable to QuecOpen® solution based on SDK build environment. It outlines Ping API, calling process and example on Quectel EC200U series and EG91xU family modules.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
-	EC200U Series
EG91xU	EG912U-GL
	EG915U Series

2 Calling Process

Before calling the Ping interface, you need to register the network and perform data call to establish the data channel. These two steps can be skipped if the data channel has been established in other tasks. The calling process is shown as below.

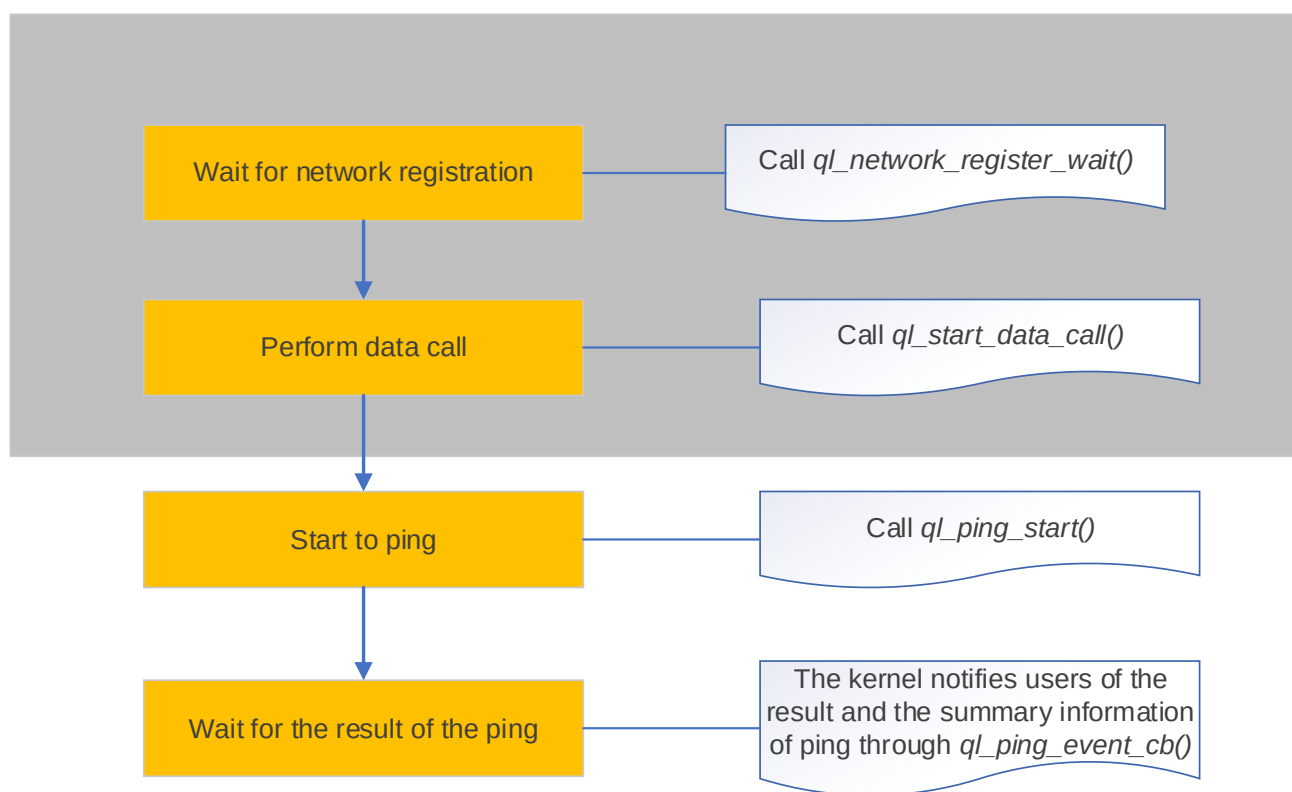


Figure 1: Ping Interface Calling Process

NOTE

See **document [2]** for details on `ql_network_register_wait()` and `ql_start_data_call()`.

3 Ping API

3.1. Header File

ql_ping_app.h, the header file of Ping API, is in the *components\ql-kernel\inc* directory. Unless otherwise specified, all the header files mentioned in this document are in this directory.

3.2. API Description

3.2.1. ql_ping_start

This function enables the ping functionality and sets the packets to be sent according to the configurations.

● Prototype

```
ping_session_id ql_ping_start(int cid, int sim_id, const char *host, ql_ping_config_type *ping_options,
ql_ping_event_cb cb_fcn)
```

● Parameter

cid:

[In] Data channel number.

sim_id:

[In] (U)SIM card number. This parameter is generally set to 0 if the module does not support dual-card function.

0 (U)SIM 1

1 (U)SIM 2

host:

[In] Remote server address pinged.

ping_options:

[In] Ping configuration. See **Chapter 3.2.1.1** for details.

cb_fcn:

[In] Callback function of ping event. See **Chapter 3.2.1.2** for details.

● Return Value

Ping session ID. See **Chapter 3.2.1.2.1** for details. Integer type. Range: 1–10. 0 indicates failed creation of a session.

3.2.1.1. ql_ping_config_type

The structure of ping configuration:

```
typedef struct
{
    uint32_t  num_data_bytes;
    uint32_t  ping_response_time_out;
    uint32_t  num_pings;
    uint32_t  ttl;
} ql_ping_config_type;
```

● Parameter

Type	Parameter	Description
uint32_t	<i>num_data_bytes</i>	Size of ping data package. Unit: byte.
uint32_t	<i>ping_response_time_out</i>	Timeout of waiting for ping response. Unit: ms.
uint32_t	<i>num_pings</i>	Number of pings.
uint32_t	<i>ttl</i>	TTL for the ping package.

3.2.1.2. ql_ping_event_cb

This is the callback function of ping event.

● Prototype

```
typedef void(*ql_ping_event_cb)(ping_session_id ping_sess_id, uint16_t event_id, int evt_code, void *evt_param)
```

● Parameter

ping_sess_id:

[In] Ping session ID. See **Chapter 3.2.1.2.1** for details.

event_id:

[In] Ping type. See **Chapter 3.2.1.2.2** for details.

evt_code:

[In] Result codes of ping. See **Chapter 3.2.1.2.3** for details.

evt_param:

[In] Parameter of ping event.

When *event_id* is *QL_PING_STATS*, see **Chapter 3.2.1.2.4** for the definition of the type structure of this parameter.

When *event_id* is *QL_PING_SUMMARY*, see **Chapter 3.2.1.2.5** for the definition of the type structure of this parameter.

- Return Value

None

3.2.1.2.1. ping_session_id

The ping session ID is defined below.

```
typedef int ping_session_id
```

3.2.1.2.2. ql_ping_event_type

The enumeration of ping type:

```
typedef enum{
    QL_PING_STATS        = 1,
    QL_PING_SUMMARY      = 2,
}ql_ping_event_type;
```

- Member

Member	Description
<i>QL_PING_STATS</i>	Result for single ping.
<i>QL_PING_SUMMARY</i>	Summary information of pings.

3.2.1.2.3. ql_ping_result_code

The enumeration of the result codes of ping:

```
typedef enum{
    QL_PING_OK                        = 0,
    QL_PING_ERR_INVALID_PARAM        = (QL_COMPONENT_LWIP_SOCKET << 16) | 552
    QL_PING_ERR_SOCKET_NEW_FAILURE   = (QL_COMPONENT_LWIP_SOCKET << 16) | 554
    QL_PING_ERR_SOCKET_BIND_FAILURE  = (QL_COMPONENT_LWIP_SOCKET << 16) | 556
    QL_PING_ERR_DNS_FAILURE           = (QL_COMPONENT_LWIP_SOCKET << 16) | 565
    QL_PING_ERR_TIMEOUT              = (QL_COMPONENT_LWIP_SOCKET << 16) | 569
}ql_ping_result_code;
```

● Member

Member	Description
<i>QL_PING_OK</i>	Successful execution.
<i>QL_PING_ERR_INVALID_PARAM</i>	Invalid parameter.
<i>QL_PING_ERR_SOCKET_NEW_FAILURE</i>	Failed to create a socket.
<i>QL_PING_ERR_SOCKET_BIND_FAILURE</i>	Failed to bind the socket.
<i>QL_PING_ERR_DNS_FAILURE</i>	Failed to parse domain name.
<i>QL_PING_ERR_TIMEOUT</i>	Execution timeout.

3.2.1.2.4. ql_ping_stats_type

The structure of single ping result:

```
typedef struct{
    uint32_t    ping_rtt;
    uint32_t    ping_size;
    uint32_t    ping_ttl;
    char        resolved_ip_addr[256];
}ql_ping_stats_type;
```

● Parameter

Type	Parameter	Description
uint32_t	<i>ping_rtt</i>	RTT value.

uint32_t	<i>ping_size</i>	Size of ping data package.
uint32_t	<i>ping_ttl</i>	TTL value for ping data package.
char	<i>resolved_ip_addr</i>	Destination address pinged.

3.2.1.2.5. ql_ping_summary_type

The structure of the summary information of pings:

```
typedef struct{
    uint32_t min_rtt;
    uint32_t max_rtt;
    uint32_t avg_rtt;
    uint32_t num_pkts_sent;
    uint32_t num_pkts_rcvd;
    uint32_t num_pkts_lost;
}ql_ping_summary_type;
```

● Parameter

Type	Parameter	Description
uint32_t	<i>min_rtt</i>	Minimum RTT. Unit: ms.
uint32_t	<i>max_rtt</i>	Maximum RTT. Unit: ms.
uint32_t	<i>avg_rtt</i>	Average RTT. Unit: ms.
uint32_t	<i>num_pkts_sent</i>	Number of data package sent.
uint32_t	<i>num_pkts_rcvd</i>	Number of response package received.
uint32_t	<i>num_pkts_lost</i>	Number of the data package lost.

4 Example

For examples of Ping functionality, see *ping_demo.c*, the Ping client demo file on the application layer, which is in the `\components\ql-application\ping` directory.

5 Appendix References

Table 2: Related Documents

Document Name
[1] Quectel_EC200U&EG91xU_Series_QuecOpen(SDK)_CSDK_Quick_Start_Guide
[2] Quectel_EC200U&EG91xU_Series_QuecOpen(SDK)_Data_Call_API_Reference_Manual

Table 3: Terms and Abbreviations

Abbreviation	Description
API	Application Programming Interface
DNS	Domain Name Server
IoT	Internet of Things
IP	Internet Protocol
Ping	Packet Internet Groper
RTOS	Real-Time Operating System
RTT	Round-Trip Time
SDK	Software Development Kit
TTL	Time to Live